

FAULT CODE 449 - Injector Metering Rail 1 Pressure - Data Valid But Above Normal Operating Range - Most Severe Level

Warnings and Cautions

WARNING

Depending on the circumstance, fuel is flammable. When performing any or all of the following procedures to remove fuel supply lines and related components, keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

CAUTION

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3823996 - female Weather-Pack™ test lead, and Part Number 3824774 - breakout cable.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Read the fault codes.	Fault Code 271, 272, 2311, 2261, 451, or 452 active or many inactive counts of fault code 271, 272, 2311, 2261, 451, or 452?
STEP 2.	Replace the mechanical dump valve.	

STEPS		SPECIFICATIONS
	STEP 2A. Check for air in fuel.	Air present in the fuel flow line?
	STEP 2B. Check the first stage fuel filter inlet pressure.	Fuel pressure greater than 0.35 bar [5 psi]?
STEP 3.	Check the operation of the fuel pump pressurizing assembly 1.	
	STEP 3A. Inspect the fuel pump pressurizing assembly 1 o-ring.	O-ring cut or shaved?
STEP 4.	Clear the fault codes.	
	STEP 4A. Disable the fault code.	Fault Code 449 inactive?
	STEP 4B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check the fault codes.

STEP 1A. Read the fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Read the fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Fault code 271, 272, 2311, 2261, 451, or 452 active or many inactive counts of fault code 271, 272, 2311, 2261, 451, or 452? YES	Appropriate troubleshooting tree.
	Fault Code 271, 272, 2311, 2261, 451, or 452 active or many inactive counts of fault code 271, 272, 2311, 2261, 451, or 452? NO	2A

STEP 2. Clear the operation of the low pressure fuel system.

STEP 2A. Check for air in fuel.

 WARNING

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system. Do not bleed the fuel system of a hot engine; this can result in fuel spilling onto a hot exhaust manifold, which can cause a fire.

Conditions :

- Remove air bleed line from air bleed valve on the fuel drain manifold block.
- Route the air bleed line into a suitable container to collect fuel.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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Inspect the fuel flow for air. Use the following procedure in Service Manual, QSK19, QSK19 CM850 Modular Common Rail System, QSK19 CM2150 Modular Common Rail System, Bulletin 4021592. Refer to Procedure 006-003 in Section 6 (/qs3/pubsys2/xml/en/procedures/20/20-006-003.html).	Air present in the fuel flow line? YES Repair : Repair or replace the damaged line or loose connection. Replace the mechanical high pressure relief valve. Use the following procedure in Service Manual, QSK19, QSK19 CM850 Modular Common Rail System, QSK19 CM2150 Modular Common Rail System, Bulletin 4021592. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html)	4A
	Air present in the fuel flow line? NO	2B

STEP 2B. Check the first stage fuel filter inlet pressure.

Conditions :

- Turn keyswitch OFF.
- Install the pressure gauge into fuel filter head port at inlet.

Action	Specification/Repair	Next Step
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Measure inlet pressure. Measure the inlet pressure to the first stage fuel filter. If the engine will not start, perform this test while cranking the engine.	Fuel pressure greater than 0.35 bar [5 psi]? YES Repair : Refer to the equipment manufacturer service information.	4A
	Fuel pressure greater than 0.35 bar [5 psi]? NO	3A

STEP 3. Check the operation of the fuel pump pressurizing assembly 1.

STEP 3A. Inspect the fuel pump pressurizing assembly 1 o-ring.

Conditions : Turn keyswitch OFF.		
Action	Specification/Repair	Next Step

Inspect the fuel pump pressurizing assembly 1. - Remove the fuel pump pressurizing assembly 1. - Use the following procedure in Service Manual, QSK19, QSK19 CM850 Modular Common Rail System, QSK19 CM2150 MCRS, Bulletin 4021592. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html) - Inspect the fuel pump pressurizing assembly 1 o-ring. - If the o-ring is cut or shaved, fuel can be bypassing the fuel pump pressurizing assembly 1 and entering the high pressure pump.	O-ring cut or shaved? YES Repair : Replace the damaged o-ring. Use the following procedure in Service Manual, QSK19, QSK19 CM850 Modular Common Rail System, QSK19 CM2150 Modular Common Rail System, Bulletin 4021592. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html)	4A
	O-ring cut or shaved? NO	4A

STEP 4. Clear the fault codes.

STEP 4A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Disable the fault code. Use INSITE™ electronic service tool to read the fault codes.	Fault Code 449 inactive? YES	4B
	Fault Code 449 inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair location if all steps have been completed and checked a second time.	1A

STEP 4B. Clear the inactive fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? YES	Repair complete
	All fault codes cleared? NO Repair : Troubleshoot any remaining active fault codes.	Appropriate troubleshooting steps